Investigating and Implementing LibreRouter and LibreMesh: Virtualized and Raspberry Pi-Based Mesh Networking

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***Abstract*—This project investigates LibreRouter and LibreMesh, open-source community networking technologies built on top of OpenWrt. The project explores the architecture, implementation, vulnerabilities, and deployment challenges associated with decentralized mesh networking systems. A virtualized OpenWrt environment was first developed using VMware Workstation Pro to simulate a four-node LibreMesh network topology. The project will later extend the deployment to Raspberry Pi hardware platforms for physical mesh networking experiments and performance evaluation.**

I. Introduction

Community mesh networks provide decentralized communication infrastructure capable of operating independently of centralized Internet Service Providers (ISPs). LibreRouter and LibreMesh were developed to support resilient, community-owned wireless infrastructure using open-source software and commodity hardware.

This project investigates the deployment and operation of LibreMesh-based networking systems using both virtualized and physical implementations. The primary goals include evaluating routing behavior, deployment complexity, security considerations, network resilience, and practical limitations of community mesh networking technologies.

In addition to traditional research analysis, this report maintains a reproducible engineering notebook style documenting implementation steps, configuration procedures, troubleshooting observations, and experimental outcomes throughout the project lifecycle.

II. Background Research

A. OpenWrt

OpenWrt is an open-source Linux-based operating system designed primarily for embedded networking devices such as routers and access points. Unlike traditional vendor firmware, OpenWrt provides a writable filesystem, package management system, and modular architecture allowing users to customize routing, firewall, wireless, and network management functionality.

The OpenWrt ecosystem supports x86 virtualized environments in addition to physical embedded hardware, making it suitable for experimental mesh networking research.

B. LibreMesh

LibreMesh is a framework built on top of OpenWrt that simplifies the deployment of decentralized mesh networking infrastructure. The platform automates many aspects of wireless mesh configuration while supporting routing protocols such as BATMAN-adv and BMX7.

LibreMesh was designed to support community-operated networks where nodes dynamically discover neighboring devices and establish resilient communication paths without centralized infrastructure.

C. LibreRouter

LibreRouter is an open-hardware networking platform developed specifically for community mesh deployments. The platform combines OpenWrt-based firmware with wireless hardware optimized for long-range and community-managed networking applications.

III. Literature Review and Security Analysis

A. Identified Weaknesses and Vulnerabilities

Several security concerns and operational limitations were identified during preliminary literature review:

- Firmware update vulnerabilities in OpenWrt package management systems
- Mesh routing attacks including route poisoning and malicious node participation
- Wireless sniffing and unauthorized network access
- Resource exhaustion and denial-of-service attacks
- Complexity of maintaining decentralized routing infrastructure

IV. Virtualized OpenWrt Deployment

A. Virtualization Environment

VMware Workstation Pro 17 was selected as the virtualization platform for the initial implementation phase. Virtualization allowed rapid deployment, rollback capability through snapshots, reproducible experimentation, and simplified troubleshooting prior to deployment on physical Raspberry Pi hardware.

B. OpenWrt Image Preparation

The OpenWrt x86_64 ext4 combined image was downloaded from the official OpenWrt release repository. Since VMware does not natively boot raw IMG files efficiently, the OpenWrt disk image was converted from IMG format into VMware VMDK format prior to deployment.

C. Virtual Machine Configuration

A custom virtual machine configuration was selected in VMware Workstation Pro to allow manual control over networking interfaces and storage configuration.

The following VM specifications were used:

TABLE I
INITIAL OPENWRT VM CONFIGURATION

Setting	Value
Guest OS	Other Linux 5.x 64-bit
vCPUs	1
Memory	256 MB
Primary Disk	OpenWrt VMDK
Network Adapter 1	NAT
Network Adapter 2	Host-only

D. Network Interface Design

Two virtual network adapters were configured for each virtual router node:

- NAT adapter for WAN connectivity and package installation
- Host-only adapter for internal mesh backbone communication

This architecture allows the virtual environment to simulate isolated mesh connectivity between router nodes while still maintaining Internet access for software installation and updates.

E. Initial Boot Process

The OpenWrt virtual machine booted successfully after the VMDK disk image was attached to the VMware virtual SCSI controller. During startup, OpenWrt successfully initialized both virtual Ethernet interfaces:

```
eth0 NIC Link is Up
eth1 NIC Link is Up
br-lan entered forwarding state
```

The successful initialization of both interfaces confirmed that VMware virtual networking devices were correctly recognized by the OpenWrt kernel.

F. Implementation Notes

- VMware generated a warning regarding a missing host CD/DVD device during startup.
- The warning was safely ignored because OpenWrt boots directly from the virtual disk image and does not require installation media after deployment.
- OpenWrt successfully booted into the root shell environment.
- The system displayed a migration notice indicating OpenWrt's transition from the legacy `opkg` package manager to the newer `apk` package management system.

V. PLANNED LIBREMESH DEPLOYMENT

The next implementation phase will involve:

- Installing LibreMesh packages
- Configuring BATMAN-adv mesh routing
- Cloning the initial VM into four router nodes
- Constructing a multi-hop virtual mesh topology
- Performing connectivity and routing validation tests

VI. FUTURE RASPBERRY PI DEPLOYMENT

The final implementation stage will migrate the virtualized mesh environment onto Raspberry Pi hardware devices running OpenWrt and LibreMesh.

The Raspberry Pi deployment will evaluate:

- Physical wireless mesh formation
- Real-world throughput and latency
- Hardware limitations
- Deployment obstacles
- Inter-group node interoperability

VII. CONCLUSION

The initial virtualization phase successfully established a functional OpenWrt router environment capable of supporting future LibreMesh experimentation. The implementation demonstrated the practicality of virtualized router deployment using VMware Workstation Pro and established the networking foundation necessary for future multi-node mesh routing experiments.

REFERENCES

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